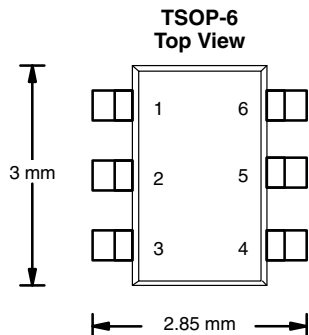


P-Channel 20-V (D-S) MOSFET

PRODUCT SUMMARY

V_{DS} (V)	$R_{DS(on)}$ (Ω)	I_D (A)	Q_g (Typ.)
- 20	0.0240 at $V_{GS} = - 4.5$ V	- 8 ^a	21 nC
	0.0372 at $V_{GS} = - 2.5$ V	- 8 ^a	



Ordering Information:

Si3407DV-T1-E3 (Lead (Pb)-free)
Si3407DV-T1-GE3 (Lead (Pb)-free and Halogen-free)

FEATURES

- Halogen-free According to IEC 61249-2-21 Definition
- TrenchFET® Power MOSFET
- PWM Optimized
- 100 % R_g Tested
- 100 % UIS Tested
- Compliant to RoHS Directive 2002/95/EC

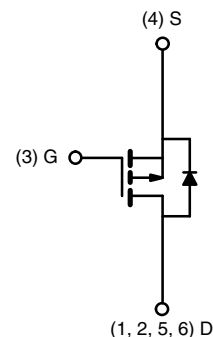
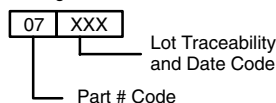


RoHS
COMPLIANT
HALOGEN
FREE
Available

APPLICATIONS

- Load Switch
- Notebook PC

Marking Code



P-Channel MOSFET

ABSOLUTE MAXIMUM RATINGS ($T_A = 25$ °C, unless otherwise noted)

Parameter	Symbol	Limit	Unit
Drain-Source Voltage	V_{DS}	- 20	V
Gate-Source Voltage	V_{GS}	± 12	
Continuous Drain Current ($T_J = 150$ °C)	I_D	- 8 ^a	A
		- 8 ^a	
		- 7.5 ^{b, c}	
		- 6 ^{b, c}	
Pulsed Drain Current	I_{DM}	- 25	
Continuous Source-Drain Diode Current	I_S	- 3.5	
		- 1.7 ^{b, c}	
Avalanche Current	I_{AS}	- 8	mJ
Single Pulse Avalanche Energy	E_{AS}	3.2	
Maximum Power Dissipation	P_D	4.2	W
		2.7	
		2 ^{b, c}	
		1.3 ^{b, c}	
Operating Junction and Storage Temperature Range	T_J, T_{stg}	- 55 to 150	°C

THERMAL RESISTANCE RATINGS

Parameter	Symbol	Typical	Maximum	Unit
Maximum Junction-to-Ambient ^{b, d}	R_{thJA}	45	62.5	°C/W
Maximum Junction-to-Foot (Drain)	R_{thJF}	25	30	

Notes:

- Package limited.
- Surface mounted on 1" x 1" FR4 board.
- $t = 5$ s.
- Maximum under steady state conditions is 110 °C/W.

SPECIFICATIONS ($T_J = 25\text{ }^{\circ}\text{C}$, unless otherwise noted)

Parameter	Symbol	Test Conditions	Min.	Typ.	Max.	Unit
Static						
Drain-Source Breakdown Voltage	V _{DS}	V _{GS} = 0 V, I _D = - 250 μA	- 20			V
V _{DS} Temperature Coefficient	ΔV _{DS} /T _J	I _D = - 250 μA		-18.7		mV/°C
V _{GS(th)} Temperature Coefficient	ΔV _{GS(th)} /T _J			3.7		
Gate-Source Threshold Voltage	V _{GS(th)}	V _{DS} = V _{GS} , I _D = - 250 μA	- 0.65		- 1.5	V
Gate-Source Leakage	I _{GSS}	V _{DS} = 0 V, V _{GS} = ± 12 V			± 100	nA
Zero Gate Voltage Drain Current	I _{DSS}	V _{DS} = - 20 V, V _{GS} = 0 V			- 1	μA
		V _{DS} = - 20 V, V _{GS} = 0 V, T _J = 55 °C			- 10	
On-State Drain Current ^a	I _{D(on)}	V _{DS} ≤ - 5 V, V _{GS} = - 4.5 V	- 25			A
Drain-Source On-State Resistance ^a	R _{DS(on)}	V _{GS} = - 4.5 V, I _D = - 7.5 A		0.0200	0.0240	Ω
		V _{GS} = - 2.5 V, I _D = - 6.4 A		0.0310	0.0327	
Forward Transconductance ^a	g _{fs}	V _{DS} = - 10 V, I _D = - 7.5 A		25		S
Dynamic ^b						
Input Capacitance	C _{iss}	V _{DS} = - 10 V, V _{GS} = 0 V, f = 1 MHz		1670		pF
Output Capacitance	C _{oss}			335		
Reverse Transfer Capacitance	C _{rss}			284		
Total Gate Charge	Q _g	V _{DS} = - 10 V, V _{GS} = - 10 V, I _D = - 7.5 A		42	63	nC
				21	32	
Gate-Source Charge	Q _{gs}	V _{DS} = - 10 V, V _{GS} = - 4.5 V, I _D = - 7.5 A		6		
Gate-Drain Charge	Q _{gd}			5		
Gate Resistance	R _g	f = 1 MHz	1.3	6.5	13	Ω
Turn-on Delay Time	t _{d(on)}	V _{DD} = - 10 V, R _L = 1.7 Ω I _D ≅ - 6 A, V _{GEN} = - 10 V, R _g = 1 Ω		8	16	ns
Rise Time	t _r			11	17	
Turn-Off Delay Time	t _{d(off)}			65	98	
Fall Time	t _f			39	59	
Turn-on Delay Time	t _{d(on)}	V _{DD} = - 10 V, R _L = 1.7 Ω I _D ≅ - 6 A, V _{GEN} = - 4.5 V, R _g = 1 Ω		32	48	
Rise Time	t _r			62	93	
Turn-Off Delay Time	t _{d(off)}			53	80	
Fall Time	t _f			38	57	
Drain-Source Body Diode Characteristics						
Continuous Source-Drain Diode Current	I _S	T _C = 25 °C			- 3.5	A
Pulse Diode Forward Current ^a	I _{SM}				- 25	
Body Diode Voltage	V _{SD}	I _S = - 6 A		- 0.8	- 1.2	V
Body Diode Reverse Recovery Time	t _{rr}	I _F = 6 A, dI/dt = 100 A/μs, T _J = 25 °C		37	56	ns
Body Diode Reverse Recovery Charge	Q _{rr}			22	33	nC
Reverse Recovery Fall Time	t _a			12		ns
Reverse Recovery Rise Time	t _b			25		

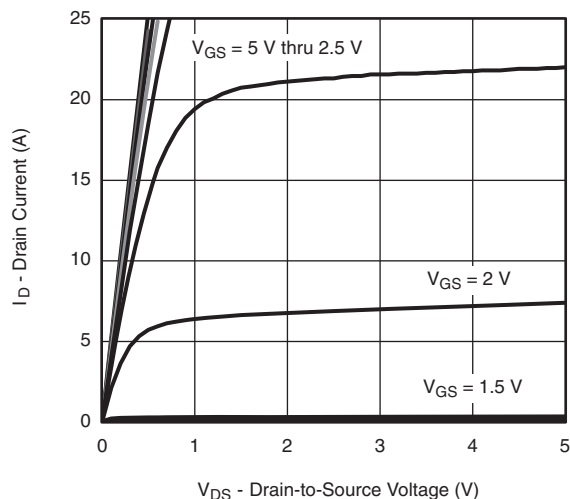
Notes:

a. Pulse test; pulse width $\leq 300\text{ }\mu\text{s}$, duty cycle $\leq 2\%$.

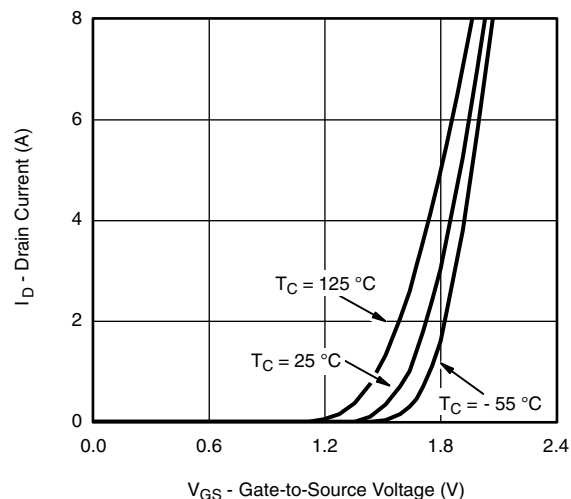
b. Guaranteed by design, not subject to production testing.

Stresses beyond those listed under "Absolute Maximum Ratings" may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated in the operational sections of the specifications is not implied. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.

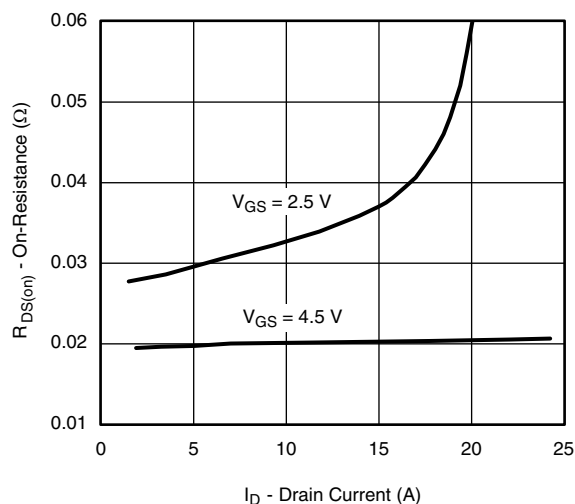
TYPICAL CHARACTERISTICS (25 °C, unless otherwise noted)



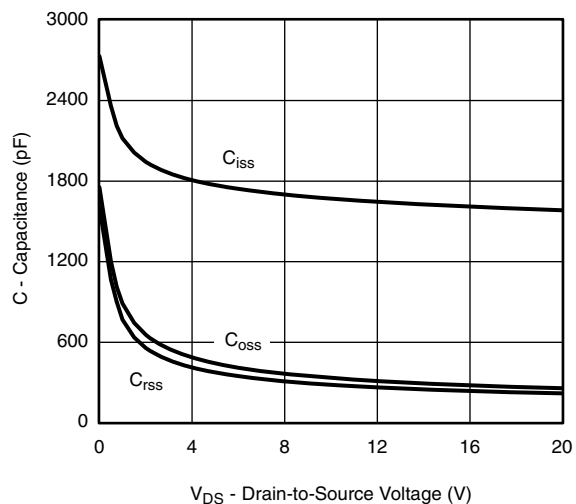
Output Characteristics



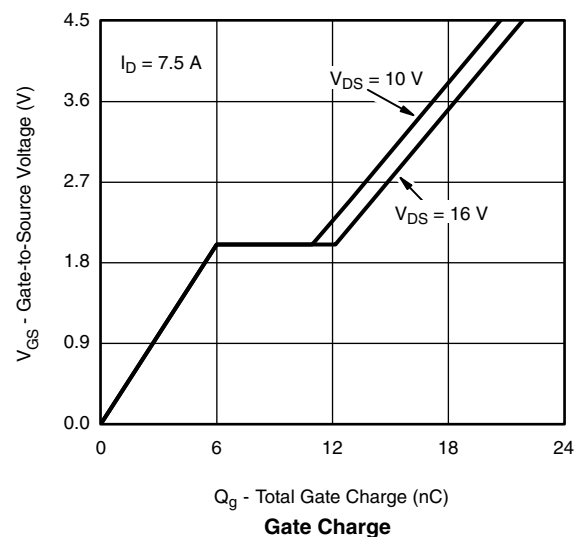
Transfer Characteristics



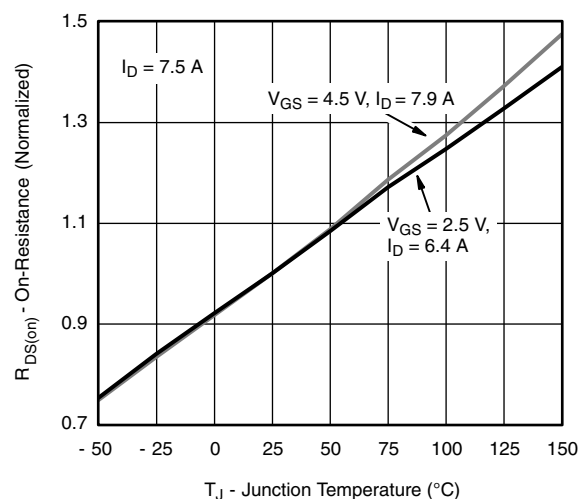
On-Resistance vs. Drain Current and Gate Voltage



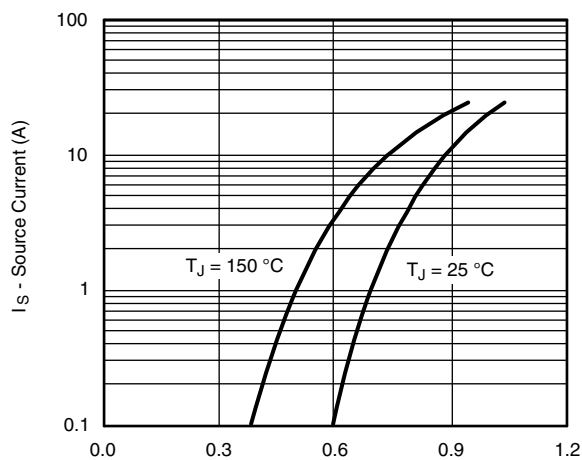
Capacitance



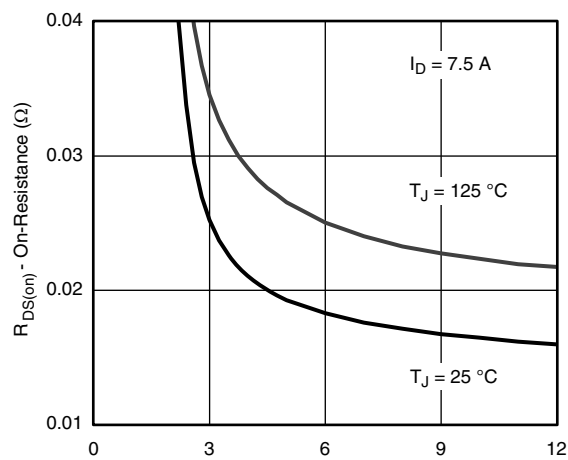
Gate Charge



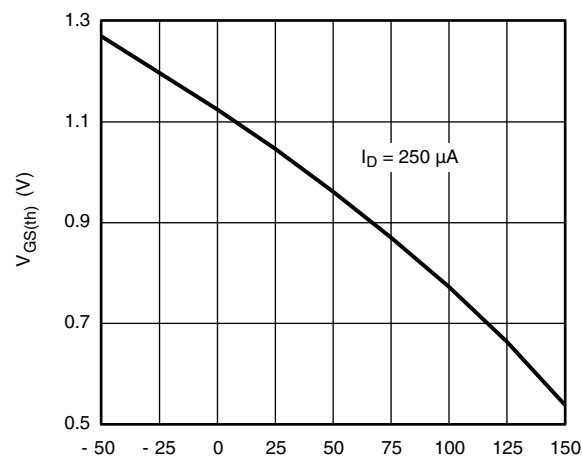
On-Resistance vs. Junction Temperature

TYPICAL CHARACTERISTICS (25 °C, unless otherwise noted)

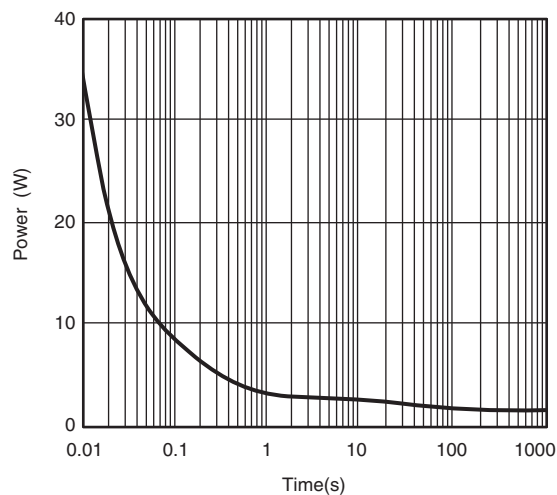
Source-Drain Diode Forward Voltage



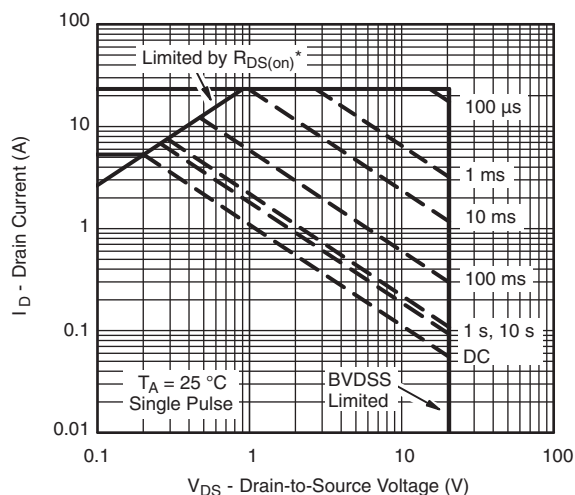
On-Resistance vs. Gate-to-Source Voltage



Threshold Voltage



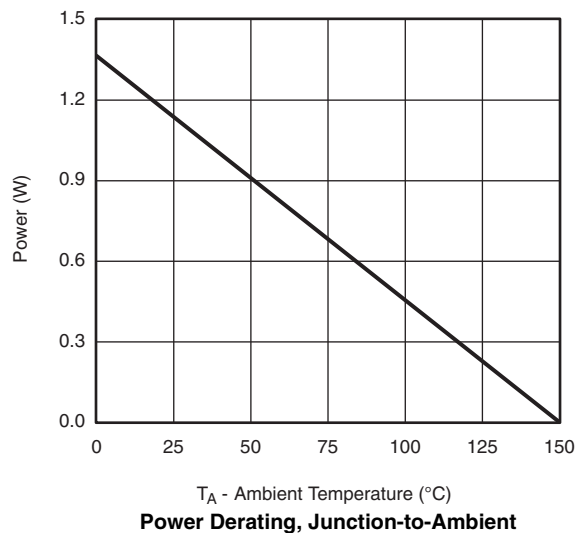
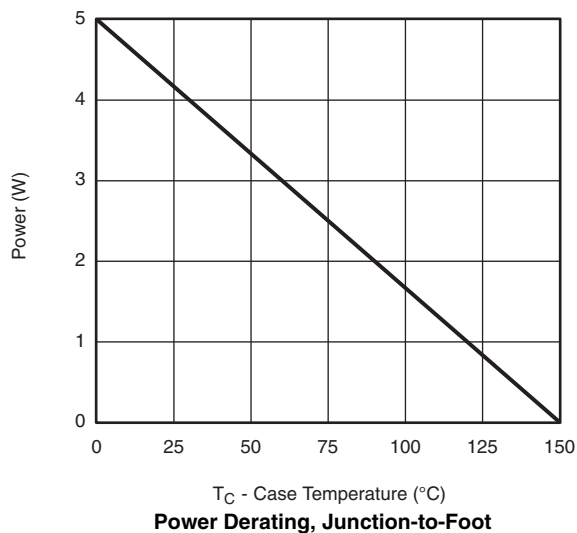
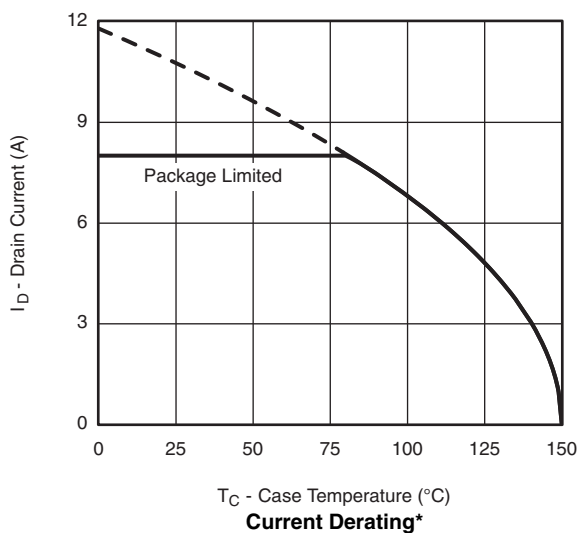
Single Pulse Power



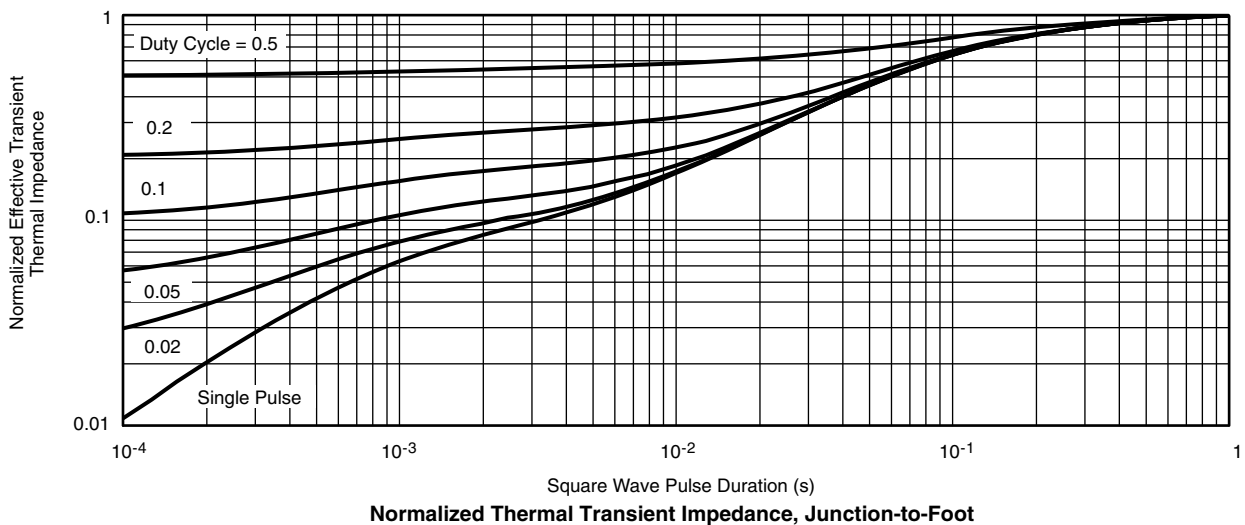
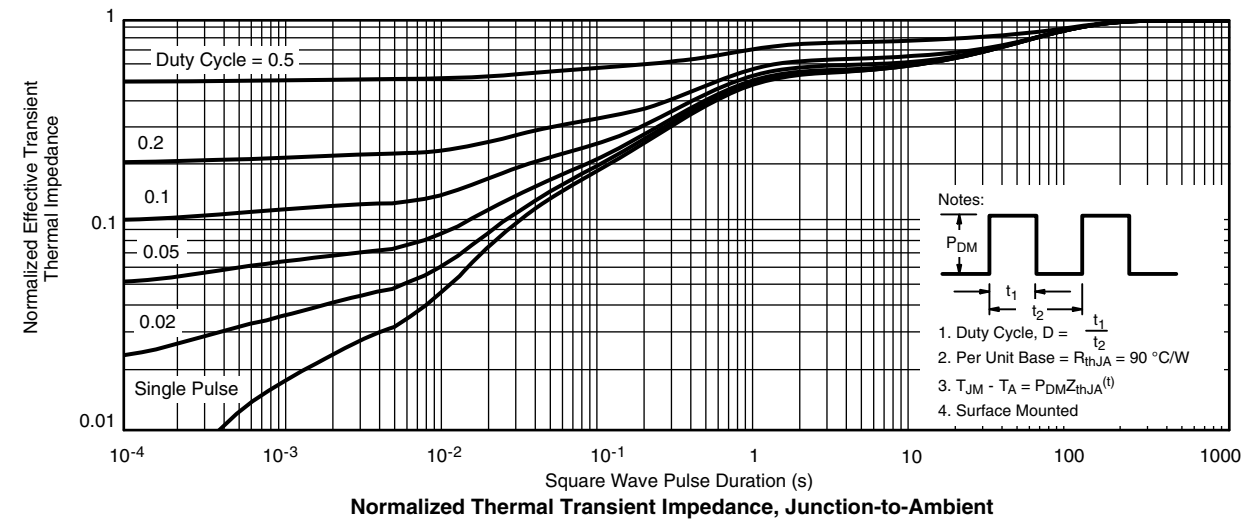
* $V_{GS} >$ minimum V_{GS} at which $R_{DS(on)}$ is specified

Safe Operating Area, Junction-to-Ambient

TYPICAL CHARACTERISTICS (25 °C, unless otherwise noted)



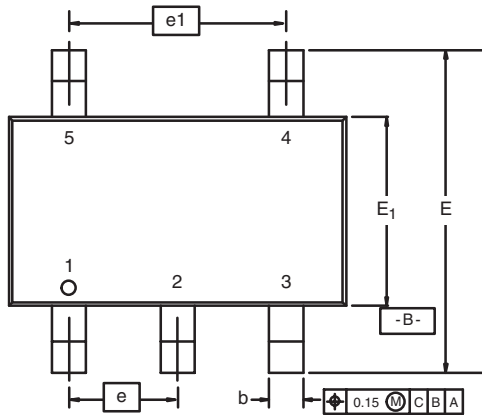
* The power dissipation P_D is based on $T_{J(max)} = 150$ °C, using junction-to-case thermal resistance, and is more useful in settling the upper dissipation limit for cases where additional heatsinking is used. It is used to determine the current rating, when this rating falls below the package limit.

TYPICAL CHARACTERISTICS (25 °C, unless otherwise noted)

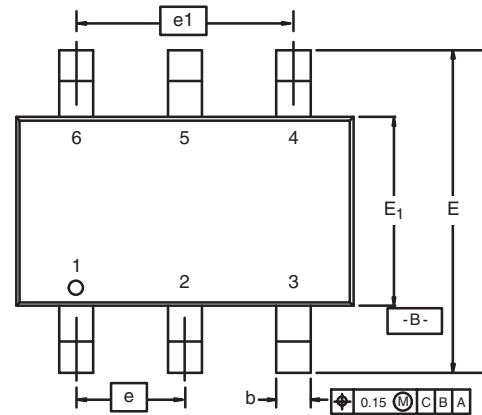
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TSOP: 5/6-LEAD

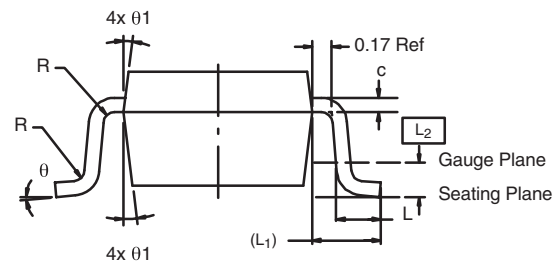
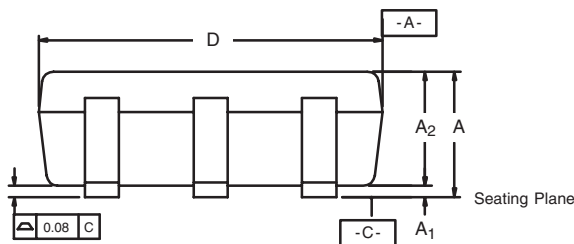
JEDEC Part Number: MO-193C



5-LEAD TSOP

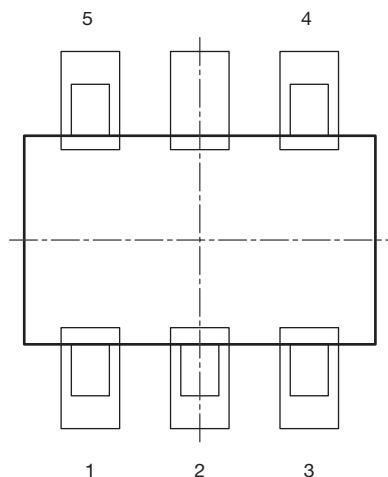


6-LEAD TSOP

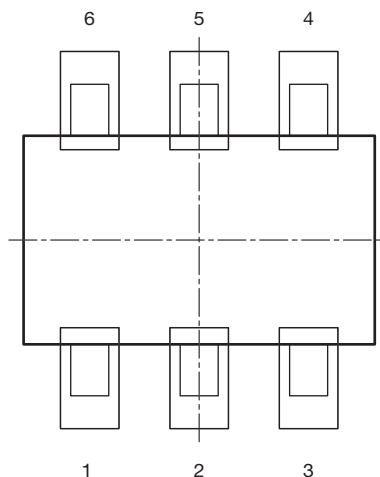


Dim	MILLIMETERS			INCHES		
	Min	Nom	Max	Min	Nom	Max
A	0.91	-	1.10	0.036	-	0.043
A ₁	0.01	-	0.10	0.0004	-	0.004
A ₂	0.90	-	1.00	0.035	0.038	0.039
b	0.30	0.32	0.45	0.012	0.013	0.018
c	0.10	0.15	0.20	0.004	0.006	0.008
D	2.95	3.05	3.10	0.116	0.120	0.122
E	2.70	2.85	2.98	0.106	0.112	0.117
E ₁	1.55	1.65	1.70	0.061	0.065	0.067
e	0.95 BSC			0.0374 BSC		
e ₁	1.80	1.90	2.00	0.071	0.075	0.079
L	0.32	-	0.50	0.012	-	0.020
L ₁	0.60 Ref			0.024 Ref		
L ₂	0.25 BSC			0.010 BSC		
R	0.10	-	-	0.004	-	-
θ	0°	4°	8°	0°	4°	8°
θ ₁	7° Nom			7° Nom		
ECN: C-06593-Rev. I, 18-Dec-06						
DWG: 5540						

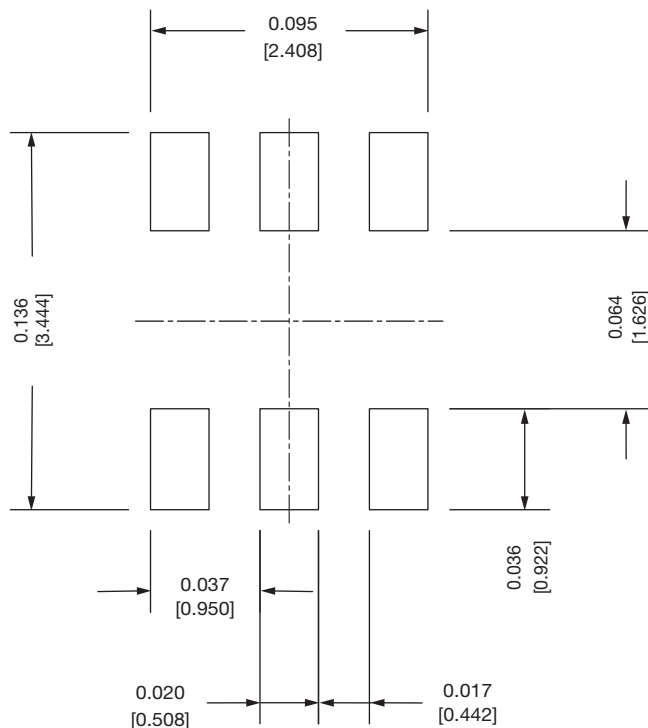
Recommended Land Pattern For TSOP-5L / TSOP-6L



TSOP 5L



TSOP 6L


Note

- All dimensions are in inches (millimeter)

ECN: C22-0860-Rev. B, 24-Oct-2022
DWG: 3010



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